

supplies, conveyor systems and access aisles.

For example, I have seen an aqueous cleaning system that used 20 square metres of floor space in the factory. But the system actually consumed 140 square metres of floor space when work-in-progress and ancillary systems were included. That is a seven times floor-space factor (140 sq.m/20 sq.m). For comparable vapour systems, which are smaller with equivalent capacity, a 4x factor would be reasonable.

To estimate floor space costs, include the cost of the space itself, plus heating, cooling, lighting, and some portion of the cost of shared facilities like the lavatories. Assuming factory space in India rents for about 1000 rupee/sq.m/month, the cost of the large aqueous cleaner cited above would be approx. 136,600 rupees per month. If the system cleans 10,000 parts per month, that adds 13.6 rupees to the cost of each part cleaned.

Fixed acquisition and installation costs. The final costs to include are the fixed and acquisition costs.

Up-front capital costs should include the costs of the machine and sub-systems, freight, site preparation and setup, building renovations, ventilation enhancements, electrical upgrades and water-treatment subsystems required to support the new system. It is essential that all of the sub-systems are included in the cost analysis.

Savvy engineers also consider making extra investments in optional features that speed cleaning or reduce costs; trading up-front capital investment for lower operating costs. For example, aqueous systems have money-saving options such as air-knives and extra drying chambers that speed cleaning but increase electrical consumption.

Planners should include the cost of the funds that will be tied up in the investment. Use the payment financial function in a spreadsheet to estimate the cost-per-month of the equipment, which can easily be converted into a cost-per-part.

Once the total purchase and installation costs are determined, divide



Fig. 4: One of the smallest vapour degreasing systems—the Unique Lab-Kleen degreaser—engineered for prototype labs, R&D and pre-production applications

that cost by the total number of parts expected to be cleaned by the system over its operational life. For example, a large 10 million Indian rupees cleaning system might be expected to last 12 years and clean 10,000 boards/month. In that case, the cost-per-part-cleaned is about 6.94 rupees.

The winning score

As we have seen, there are many factors to tabulate when selecting a new cleaning system. A detailed spreadsheet is available for downloading (from the website 'http://www.microcareprecisioncleaners.com/assets/documents/uploads/Degreaser_Costs_Compared.xls') that could serve as a template for engineers planning this comparison. But, in general, use the following cleaning scoreboard to find the winning technology for your company:

1. Determine the likely cleaning requirements for today's products as well as those of tomorrow. Average those requirements into an hourly rate of required throughput.

2. Compare different cleaning technologies. Send samples to the equipment makers to prove the ability of their systems to clean the components to your specifications.

3. From among the remaining candidate systems, collect comparative data on every important characteristic. Be sure to include up-front capital costs, plus the direct and indirect operational costs (supervisory and engineering labour costs, plus energy, water, solvent, labour and maintenance costs).

4. As the various costs are collected, divide them by the expected throughput of the system to calibrate them into a cost-per-part-cleaned index.

5. Sum all the data into a single performance index, the total cost-per-part-cleaned.

6. Select the system with the lowest total cost-per-part-cleaned.

Using standard statistical tools, engineers can model all the operating costs for systems of different types and sizes. If this process is completed accurately, thoroughly and impartially, the savvy engineer can be confident that the selected system will become a valued part of the production process for years to come. ●

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